

Prelab 6.1 Database & Visualization 1 - SQL Database

Learning Goals

Students will be able to:

1. Execute basic SQL statements and clauses to perform actions on a database
2. Write a Python script to execute SQL statements that modify a SQL database

1.1 Introduction

In this lab, we will use a database to store data that is collected by IoT sensors and devices. In Lab 4 and 5, we learned MTConnect as a middleware to integrate data flow. Because data size in the MTConnect agent is limited to the memory size of the computer, introducing database is needed to store data for long-term use.

There are many different types of databases. Normally the types of databases are divided into SQL (Structured Query Language) and NoSQL (non SQL or not only SQL). The selection of a proper database is important when you start a project. Of course, NoSQL can be better in many applications. However, of the various databases, we will use SQL (Structured Query Language) database, specifically MySQL. SQL is a standardized programming "language" that is used to manage relational databases and perform various operations on the data in them. Because SQL database is elementary of database, we will use SQL for the lab activities. SQL allow us to store, update, and retrieve data in databases. MySQL is an open-source (royalty free) relational database management system (RDBMS). It is developed by Oracle Co. MySQL is a representative RDBMS and it is used by many popular websites, including Facebook, Flickr, YouTube and so on.

The purpose of Lab 7 is 1) to understand the basic syntax of SQL and 2) to manipulate (store and retrieve) databases. As a hands-on activity, we will store sensor data into MySQL database using SQL and retrieve data from it.

1.2 MySQL Python Package Installation on Raspberry Pi

In Lab 6, we will program a Python script to store sensor data into MySQL database. PyMySQL (<https://pypi.org/project/PyMySQL/>, available on Feb. 28, 2022) will be used. Please follow the steps to install PyMySQL Python package below.

1. To make your Raspberry Pi up to date and upgrade



Raspberry Pi - Terminal

```
sudo apt update
```

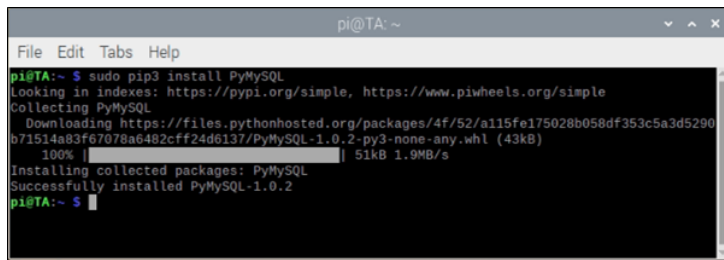
```
sudo apt upgrade
```

2. Install PyMySQL Python package (latest version is 1.0.2)



Raspberry Pi - Terminal

```
sudo pip3 install PyMySQL
```

A terminal window titled 'pi@TA: ~' showing the installation of PyMySQL. The user runs 'sudo pip3 install PyMySQL'. The terminal output shows the package being collected, downloaded from a URL, and successfully installed as PyMySQL-1.0.2.

```
pi@TA:~ $ sudo pip3 install PyMySQL
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Collecting PyMySQL
  Downloading https://files.pythonhosted.org/packages/4f/52/a115fe175028b058df353c5a3d5290b71514a83f67078a6482cff24d6137/PyMySQL-1.0.2-py3-none-any.whl (43kB)
    100% |#####| 51kB 1.9MB/s
Installing collected packages: PyMySQL
Successfully installed PyMySQL-1.0.2
pi@TA:~ $
```

Figure 1 PyMySQL installation on Raspberry Pi

3. Check if PyMySQL is installed as below

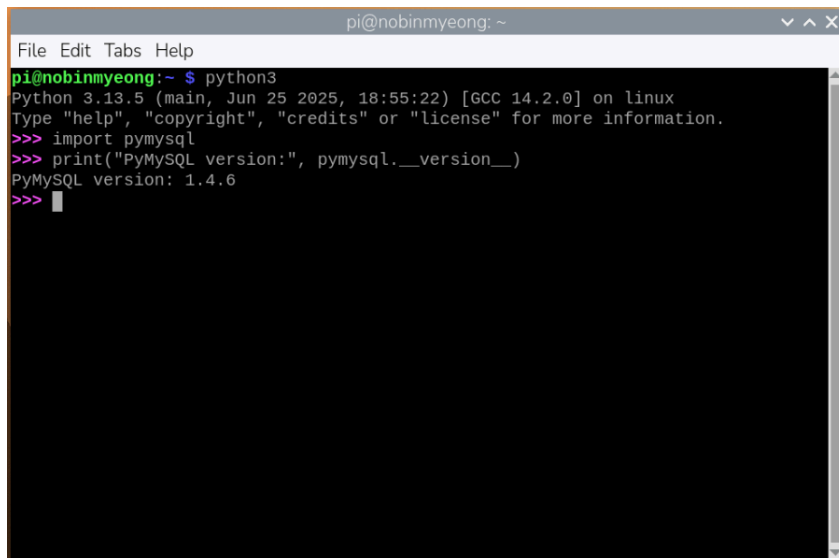
Raspberry Pi - Terminal

```
python3
```

Python - Python3 interpreter in 'Terminal'

```
import pymysql
```

```
print("PyMySQL version:", pymysql.__version__)
```

A terminal window titled 'pi@noblinmyeong: ~' showing the Python3 interpreter. The user runs 'python3', which shows the Python version and environment. Then, the user enters 'import pymysql' and 'print("PyMySQL version:", pymysql.__version__)', which outputs 'PyMySQL version: 1.4.6'.

```
pi@noblinmyeong:~ $ python3
Python 3.13.5 (main, Jun 25 2025, 18:55:22) [GCC 14.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import pymysql
>>> print("PyMySQL version:", pymysql.__version__)
PyMySQL version: 1.4.6
>>>
```

Figure 2 PyMySQL import and check the version

Task 1.1

Install PyMySQL package on Raspberry Pi. Check the version of PyMySQL and then attach the result of the terminal window as Figure 2 to the report.

```
pi@robertclaud: ~/Documents/lab
(lab-venv) pi@robertclaud:~/Documents/lab $ python
Python 3.13.5 (main, Jun 25 2025, 18:55:22) [GCC 14.2.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>> import pymysql
>>> print("PyMySQL version:", pymysql.__version__)
PyMySQL version: 1.4.6
>>>
```

- To exit the Python interpreter, the command is 'exit()'.

1.3 MySQL Workbench Installation on Windows

We can manage MySQL database through command line interface. However, it requires fluent skills on both MySQL client and CLI. We will use [MySQL Workbench](#). MySQL Workbench is a unified visual tool for database architects, developers. It provides data modeling, SQL development, and comprehensive administration tools. We will use MySQL Workbench to practice basic syntax of SQL. To install MySQL Workbench, please follow the steps below.

1. Visit [MySQL Community Downloads](#)
 2. Download and install MySQL Workbench Community version 8.0.X
- a. You can install other versions but please make sure to install the version higher than 8.0.

🔗 [MySQL Community Downloads](#)

⬅ [MySQL Workbench](#)

[General Availability \(GA\) Releases](#) [Archives](#) [Download](#)

MySQL Workbench 8.0.45

Select Operating System:

Recommended Download:

MySQL Installer for Windows

All MySQL Products. For All Windows Platforms. In One Package.

Starting with MySQL 5.6 the MySQL Installer package replaces the standalone MSI packages.

Windows (x86, 32 & 64-bit), MySQL Installer MSI

[Go to Download Page >](#)



Other Downloads:

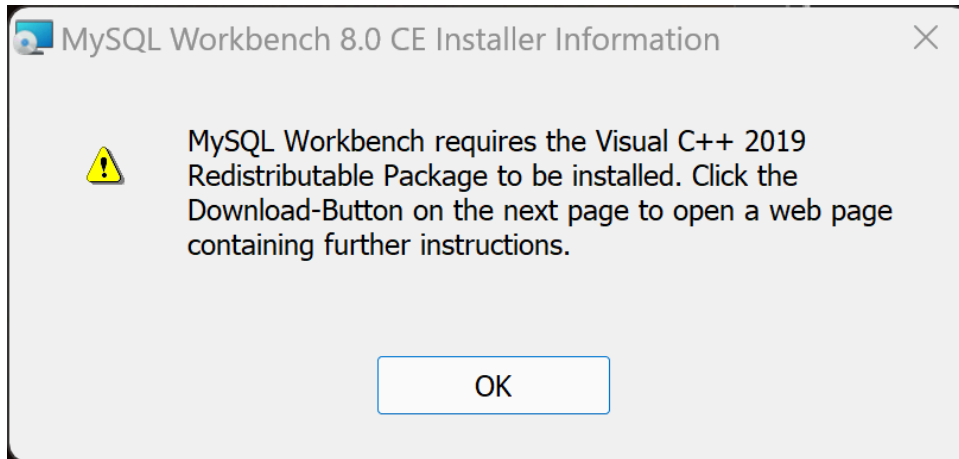
Windows (x86, 64-bit), MSI Installer	8.0.45	252.3M	Download
(mysql-workbench-community-8.0.45-win64.msi)			
MD5: 7374733ebabd22f1240bc18d936b4eb8 Signature			

🔔 We suggest that you use the [MD5 checksums](#) and [GnuPG signatures](#) to verify the integrity of the packages you download.

Figure 3 MySQL Workbench window

Troubleshooting

If you encounter an error message while downloading MySQL, it means that the compatible 'Visual C++' software is not installed. Please follow the guidelines and retry the download process.



First, enter the following Visual C++ download link.

<https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist?view=msvc-170>

Click the following link, as shown in the red box.

[Learn / C++, C, and Assembler /](#)

[v](#) [+](#) [p](#) [:](#)

Microsoft Visual C++ Redistributable latest supported downloads

Article • 11/15/2024 • 6 contributors

[Feedback](#)

In this article

[Visual Studio 2015, 2017, 2019, and 2022](#)

[Latest Microsoft Visual C++ Redistributable Version](#)

[Visual Studio 2013 \(VC++ 12.0\) \(no longer supported\)](#)

[Visual Studio 2012 \(VC++ 11.0\) Update 4 \(no longer supported\)](#)

[Show 5 more](#)

Select a C++ download link that matches your OS version and is compatible.

(x86: Window 32-bit, x64: Window 64-bit, ARM64: Linux 64-bit)

Latest Microsoft Visual C++ Redistributable Version

The latest version is `14.42.34433.0`

Use the following links to download this version for each supported architecture:

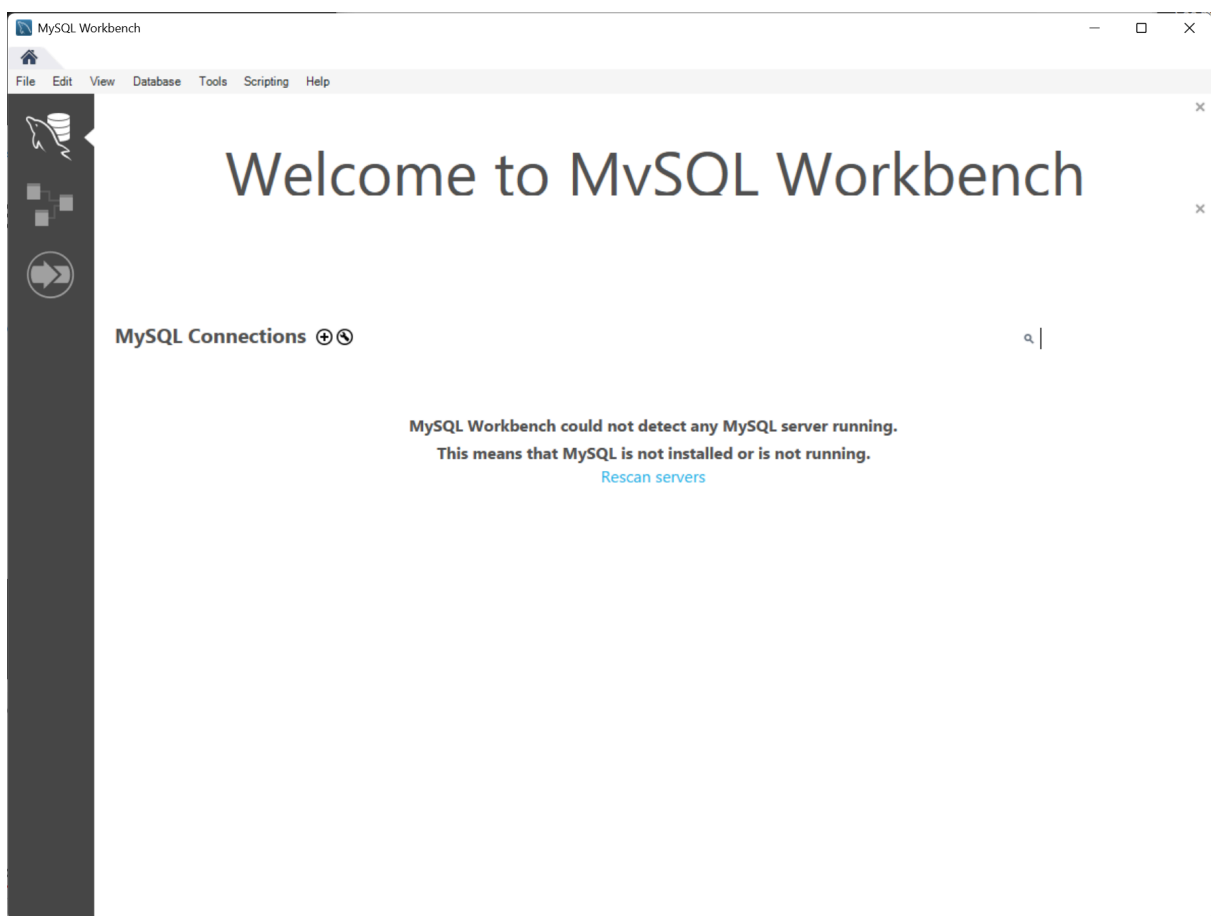
[Expand table](#)

Architecture	Link	Notes
ARM64	https://aka.ms/vs/17/release/vc_redist.arm64.exe	Permalink for latest supported ARM64 version
X86	https://aka.ms/vs/17/release/vc_redist.x86.exe	Permalink for latest supported x86 version
X64	https://aka.ms/vs/17/release/vc_redist.x64.exe	Permalink for latest supported x64 version. The X64 Redistributable package contains both ARM64 and X64 binaries. This package makes it easy to install required Visual C++ ARM64 binaries when the X64 Redistributable is installed on an ARM64 device.

Download other versions, including long term servicing release channel (LTSC) versions, from my.visualstudio.com.

Task 1.2

Install MySQL Workbench on Windows and run it. Capture the window as Figure 3 and then attach it to the report.



Task 1.3

Complete the SQL tutorial in the following URL from the section 'SQL Home' to the section 'SQL Like'.

- (Tutorial) W3Schools: <https://www.w3schools.com/sql/default.asp>

Task 1.4 Answer the following questions

1. Describe the structure of a table and the components that make up a SQL table.

A table is defined by a name and contains records or rows of data. SQL tables are formatted similarly to excel or .csv tables in that they are tabular in the format of columns and rows. SQL tables differ in that tables can be relational to one another. Several tables can all relate to one another even when they are different sizes and contain different data.

2. Explain the function of each of the SQL statements or clauses below.

T4_SELECT: SELECT is the statement in which you choose which column to pull data from. T4_INSERT_INTO: INSERT_INTO is the statement used to add more data to a table in a SQL database. T4_WHERE: WHERE is a clause used to add conditionals to filter out certain parts of the data retrieved from the database. T4_ORDER_BY: ORDER_BY is a clause used to sort the data by one of the columns in a database in ascending or descending order. T4_FROM: FROM is the statement in which you declare the table to pull the selected columns from.

Get back to [Lab Index Page](#)